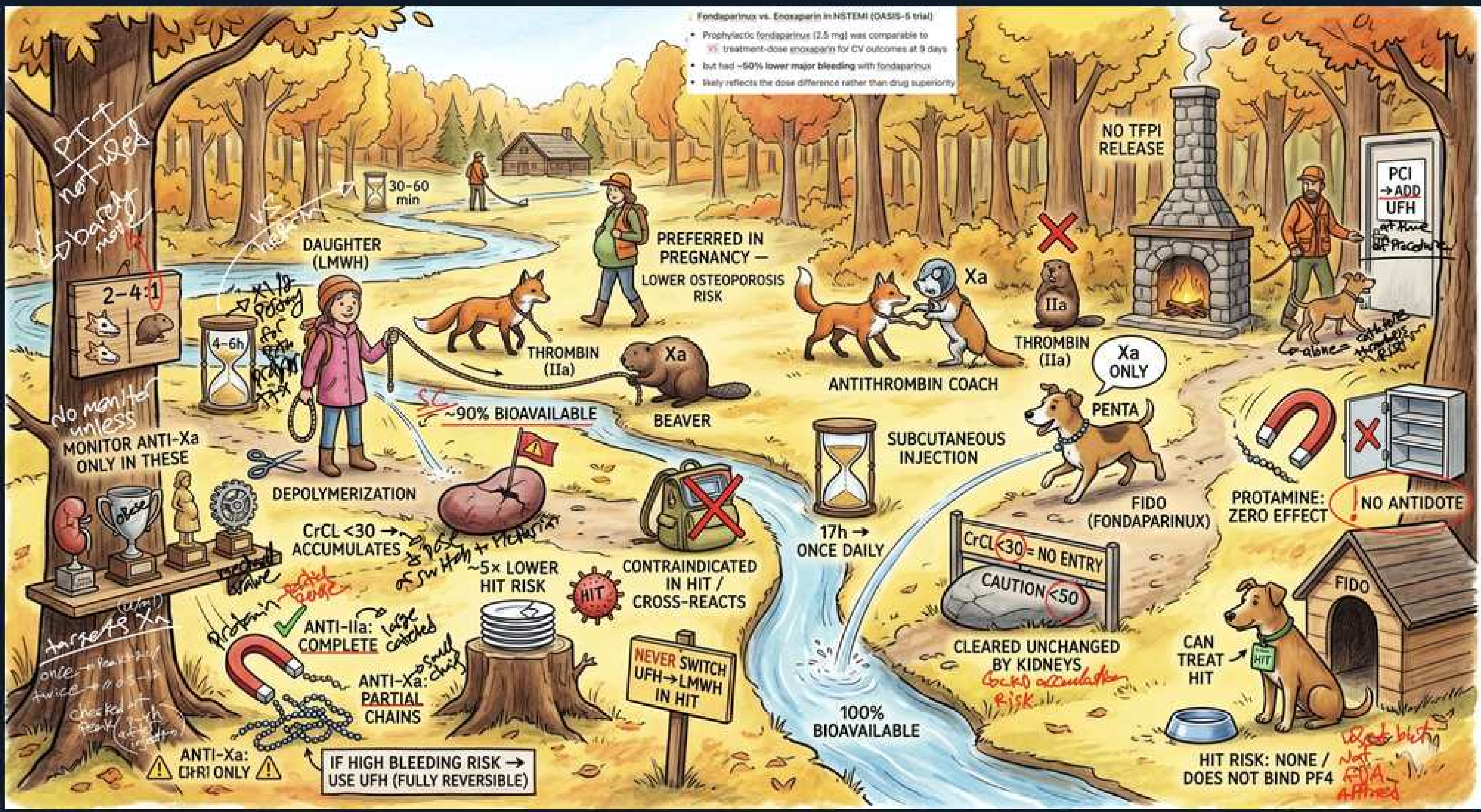
Parenteral — LMWH & Fondaparinux Forest



1. LMWH (Daughter) — depolymerized heparin

WHAT YOU SEE

Child figure 'DAUGHTER (LMWH)', 'DEPOLYMERIZATION'

WHAT IT ENCODES

Low-molecular-weight heparins (enoxaparin, dalteparin, tinzaparin) are UFH that has been DEPOLYMERIZED into SHORT chains. Like UFH they act via antithrombin, but the shorter chains preferentially inhibit factor Xa over thrombin (anti-Xa:anti-IIa ≈ 2:1 to 4:1). Given subcutaneously, once or twice daily, with predictable kinetics.

BOARD TRAP

LMWH is mainly an anti-Xa agent (most chains too short to bridge antithrombin to thrombin) — UFH inhibits IIa and Xa equally (1:1). Chain length sets the anti-Xa:anti-IIa ratio.

2. Anti-Xa : anti-IIa ratio 2–4:1

WHAT YOU SEE

Top-left scales/note '2–4:1'

WHAT IT ENCODES

The shorter LMWH chains retain the antithrombin-binding pentasaccharide (so anti-Xa activity is preserved) but most are too short (<18 saccharides) to bridge thrombin — giving a 2–4:1 anti-Xa:anti-IIa ratio. Fondaparinux, a pure pentasaccharide, is even more selective: PURE anti-Xa, no anti-IIa.

BOARD TRAP

Ordering of selectivity: UFH (1:1) < LMWH (2–4:1) < fondaparinux (pure anti-Xa). The shorter the chain, the more Xa-selective and the less thrombin inhibition.

3. 90%+ bioavailability / once-daily

WHAT YOU SEE

Beaver + 'Xa', '~90% BIOAVAILABLE', hourglass '17h once daily' for fondaparinux

WHAT IT ENCODES

LMWH has HIGH (~90%) and predictable subcutaneous bioavailability with LINEAR, dose-independent kinetics — that is why it needs NO routine monitoring and is dosed by weight. Fondaparinux has a long half-life (~17 h) allowing ONCE-DAILY dosing.

BOARD TRAP

LMWH/fondaparinux have predictable kinetics and need no routine aPTT — UFH (poor, saturable SC bioavailability) does. Don't monitor LMWH with aPTT; if a level is needed use anti-Xa.

4. Monitor anti-Xa only in special cases

WHAT YOU SEE

'NO MONITORING unless...', 'MONITOR ANTI-Xa ONLY IN THESE'

WHAT IT ENCODES

Routine LMWH monitoring is unnecessary, but an ANTI-Xa level (LMWH-calibrated, drawn ~4 h post-dose at peak) is warranted in SPECIAL populations: pregnancy, obesity/extremes of weight, renal impairment, and pediatrics — where weight-based dosing may not predict exposure reliably.

BOARD TRAP

LMWH is NOT monitored by aPTT — only by anti-Xa, and only in special cases (pregnancy, obesity, renal failure, children). Choosing aPTT to monitor LMWH is wrong.

5. CrCl <30 accumulates — caution

WHAT YOU SEE

Fondaparinux dog 'FIDO', 'CrCl <30 → NO ENTRY', 'CAUTION <50', 'CLEARED UNCHANGED BY KIDNEYS'

WHAT IT ENCODES

Both LMWH and fondaparinux are RENALLY cleared (fondaparinux is excreted essentially UNCHANGED), so they ACCUMULATE in renal impairment and raise bleeding risk. LMWH should be dose-reduced/anti-Xa-monitored at CrCl <30; FONDAPARINUX is CONTRAINDICATED at CrCl <30 and used with caution <50.

BOARD TRAP

In severe renal failure (CrCl <30) prefer UFH (extra-renal clearance, protamine-reversible). Fondaparinux is outright contraindicated <30; full-dose LMWH accumulates and bleeds. This renal pivot is high-yield.

6. Lower heparin-induced thrombocytopenia risk

WHAT YOU SEE

'LOWER HIT RISK', red marks

WHAT IT ENCODES

LMWH causes the immune heparin-induced thrombocytopenia reaction MUCH LESS often than UFH (shorter chains bind PF4 less), but the risk is NOT zero. Fondaparinux essentially does NOT cause it and does not bind PF4 — it is sometimes used to treat it (though not formally first-line).

BOARD TRAP

LMWH still cross-reacts with heparin-induced-thrombocytopenia antibodies, so it is CONTRAINDICATED once that diagnosis is established — switching UFH→LMWH in that setting is wrong. Fondaparinux does not bind PF4 (HIT risk ~none).

7. Never switch UFH→LMWH in this reaction

WHAT YOU SEE

Sign 'CONTRAINDICATED IN HIT / CROSS-REACTS', 'NEVER SWITCH UFH→LMWH IN HIT'

WHAT IT ENCODES

Because LMWH CROSS-REACTS with the same heparin–PF4 antibodies, it cannot be used once heparin-induced thrombocytopenia is suspected/confirmed. Stop ALL heparin products and start a non-heparin anticoagulant — a direct thrombin inhibitor (argatroban, bivalirudin) or fondaparinux.

BOARD TRAP

The correct switch in heparin-induced thrombocytopenia is to a direct thrombin inhibitor (argatroban — hepatically cleared, good in renal failure; bivalirudin), NOT to LMWH. Argatroban is preferred when renal function is poor.

8. Protamine only partial / zero reversal

WHAT YOU SEE

Magnet 'PROTAMINE', 'NO ANTIDOTE' for fondaparinux, 'ZERO EFFECT'

WHAT IT ENCODES

PROTAMINE only PARTIALLY reverses LMWH (~60% of anti-Xa activity, because it cannot bind the shortest chains). Protamine has essentially ZERO effect on FONDAPARINUX — there is NO specific antidote (andexanet or rFVIIa are off-label salvage options).

BOARD TRAP

Reversibility ranking: UFH fully reversible by protamine > LMWH only partial > fondaparinux essentially NONE. Don't expect protamine to fix a fondaparinux bleed.

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9. Preferred in pregnancy

WHAT YOU SEE

Top banner 'PREFERRED IN PREGNANCY — LOWER OSTEOPOROSIS RISK'

WHAT IT ENCODES

LMWH is the PREFERRED anticoagulant in PREGNANCY: it does not cross the placenta and, versus UFH, carries LOWER risk of OSTEOPOROSIS and heparin-induced thrombocytopenia, with convenient SC dosing. Fondaparinux is reserved for those who cannot take heparins (e.g., heparin-induced thrombocytopenia or heparin allergy).

BOARD TRAP

Pregnancy anticoagulation = LMWH (not warfarin — teratogenic; not DOACs — insufficient data). UFH is used peripartum for its short, reversible action, but LMWH is preferred during the pregnancy itself.

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10. Antithrombin coach (mechanism)

WHAT YOU SEE

Figure 'ANTITHROMBIN COACH', 'Xa', 'IIa' targets

WHAT IT ENCODES

All three (UFH, LMWH, fondaparinux) are INDIRECT anticoagulants that require ANTITHROMBIN as a cofactor — they accelerate antithrombin's inactivation of clotting factors rather than acting directly. Fondaparinux's pure pentasaccharide drives antithrombin to inhibit Xa ONLY.

BOARD TRAP

In antithrombin DEFICIENCY all heparins/fondaparinux lose efficacy ('resistance') — the answer is antithrombin repletion or a DIRECT inhibitor (argatroban/bivalirudin/DOAC), which act independently of antithrombin.

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11. Subcutaneous injection

WHAT YOU SEE

Fido leaping, 'SUBCUTANEOUS INJECTION', '100% BIOAVAILABLE'

WHAT IT ENCODES

LMWH and fondaparinux are given by SUBCUTANEOUS injection (fondaparinux ~100% bioavailable SC), enabling outpatient/home anticoagulation without an IV drip or routine labs — a major practical advantage over IV UFH for stable VTE and bridging.

BOARD TRAP

UFH for full anticoagulation is typically IV (continuous, monitored); LMWH/fondaparinux are SC with predictable absorption and no routine monitoring — a key administration distinction.

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12. Fido = fondaparinux (pure anti-Xa)

WHAT YOU SEE

Dog 'FIDO (FONDAPARINUX)', 'Xa ONLY', 'PENTA'

WHAT IT ENCODES

FONDAPARINUX ('Fido') is a synthetic PENTASACCHARIDE — the minimal antithrombin-binding sequence — giving PURE anti-Xa activity and NO anti-IIa. Long half-life (~17 h) means once-daily SC dosing; cleared unchanged by kidneys; no PF4 binding so negligible heparin-induced-thrombocytopenia risk.

BOARD TRAP

Fondaparinux is pure anti-Xa (no thrombin inhibition), has NO reversal agent, is contraindicated at CrCl <30, and is the one heparin-class drug essentially free of heparin-induced thrombocytopenia (and even usable to treat it).

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13. HIT risk: none / does not bind PF4

WHAT YOU SEE

Bottom-right 'HIT RISK: NONE / DOES NOT BIND PF4', Fido

WHAT IT ENCODES

Because fondaparinux does NOT bind platelet factor 4, it does not form the antigenic heparin–PF4 complex and so carries essentially NO risk of heparin-induced thrombocytopenia — and is sometimes used off-label to treat it when a direct thrombin inhibitor is unsuitable.

BOARD TRAP

Risk ranking for heparin-induced thrombocytopenia: UFH (highest) > LMWH (much lower but real, cross-reacts) > fondaparinux (essentially none). Don't claim LMWH is 'safe' in established heparin-induced thrombocytopenia — it cross-reacts.

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14. Use UFH if high bleeding risk (reversible)

WHAT YOU SEE

Bottom-left 'IF HIGH BLEEDING RISK → USE UFH (FULLY REVERSIBLE)'

WHAT IT ENCODES

When bleeding risk is high or the patient may need urgent surgery, choose UFH over LMWH/fondaparinux: UFH has a short half-life and is FULLY reversible with protamine, whereas LMWH is only partially reversible and fondaparinux is not reversible at all.

BOARD TRAP

High bleeding risk / imminent procedure / severe renal failure all favor UFH (short, reversible, extra-renal). LMWH and fondaparinux are for stable outpatients with adequate renal function.
